Pre-Systems Lesson
Two variable systems to introduce three variable systems

Solving systems of two linear equations using substitution

2 × 2 Substitution	
Key Concepts	Substitution is a method for solving systems of linear equations that involves solving one equation for a specific variable and substituting the new expression into the other equation(s) with the goal of eliminating a variable. 2 × 2 systems of linear equations have two variables and two equations. The solution to a two variable system of equations may be represented by an ordered pair. when there is one solution
Process	**Steps to Solve 2 × 2 Systems of Linear Equations using Substitution** 1. Isolate a variable in one equation. 2. Substitute the expression representing that variable into the other equation. 3. Simplify and solve. 4. Substitute the value found in step 3 into either of the original equations to solve for the remaining variable. 5. Write the solution as an ordered pair, if appropriate.

1. $-3x + 7y = 14$
   $2x - y = -13$

$2x - y = -13$
$-2x \quad -2x$
$-y = -13 - 2x$
$y = 13 + 2x$
$y = 13 + 2(-7)$
$y = 13 - 14 = -1$

$-3x + 7(13 + 2x) = 14$
$-3x + 91 + 14x = 14$
$11x + 91 = 14$
$11x = -77$
$x = -7$

$\boxed{(-7, -1)}$

2. $x + y = 16$
   $2y = -2x + 2$

$x = 16 - y$
$2y = -2(16 - y) + 2$
$2y = -32 + 2y + 2$
$2y = -30 + 2y$
$-2y \quad -2y$
$0 = -30$ $\boxed{\text{No soln.}}$

3. $2b = 6a - 14$
   $3a - b = 7$

$-b = 7 - 3a$
$b = -7 + 3a$
$2(-7 + 3a) = 6a - 14$
$-14 + 6a = 6a - 14$
$0 = 0$
$\boxed{\text{inf. soln.}}$

4. Sam sells athletic shoes part-time at a department store. He can earn either $500 per month plus 4% commission on his total sales, or $400 per month plus a 5% commission on total sales. Write a system of equations to represent this situation and find the total price of the athletic shoes he needs to sell to earn the same income from each pay scale.

$y = .04x + 500$
$y = .05x + 400$

$.05x + 400 = .04x + 500$
$-.04x \quad -400 \quad -.04x \quad -400$
$.01x = 100$
$\boxed{x = 10,000}$

Solving systems of linear equations using <u>elimination</u>

2 × 2 Elimination (Combination)	
★	Elimination is a method for solving systems of linear equations that involves adding the multiple of one equation to the multiple of another equation with the goal of eliminating a variable.
Process	**Steps to Solve 2 × 2 Systems of Linear Equations by Elimination (Combination)** 1. Multiply one or both of the equations by a constant to obtain coefficients that differ only in sign for one of the variables. 2. Add the revised equations, with the goal of eliminating one of the variables. 3. Simplify and solve for the remaining variable. 4. Substitute the value found in step 3 into either of the original equations to solve for the remaining variable. 5. Write the solution as an ordered pair, if appropriate.

5. $(3x - y = 10)3$
$4x + 3y = -4$

$9x - 3y = 30$
$4x + 3y = -4$
$13x = 26$
$x = 2$
$3(2) - y = 10$
$6 - y = 10$
$-y = 4$
$y = -4$
$\boxed{(2,-4)}$

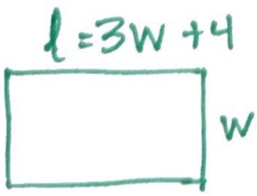
6. $(3x + 6y = 11)-2$
$(2x + 4y = 9)3$

$-6x - 12y = -22$
$6x + 12y = 27$
$0 = 5$
$\boxed{\text{No soln.}}$

7. $(x - 2y = 2)-3$
$3x - 6y = 6$

$-3x + 6y = -6$
$3x - 6y = 6$
$0 = 0$
$\boxed{\text{Inf. many sol.}}$

8. The length of Sally's garden is 4 meters greater than 3 times the width. The perimeter of her garden is 72 meters. What are the dimensions of Sally's garden?



$2l + 2w = 72$
$l = 3w + 4$

$2l + 2w = 72$
$-2(l - 3w = 4)$

$2l + 2w = 72$
$-2l + 6w = -8$
$8w = 64$

$\boxed{w = 8\text{ m}}$
$l = 3(8) + 4$
$l = 24 + 4$
$\boxed{l = 28\text{ m}}$